



TECHNOVA MURDER MYSTERY

Major Project for 21 Days SQL Challenge
by Indian Data Club

CONFIDENTIAL INVESTIGATION FILE #8842

Mission Briefing

THE INCIDENT

Date: October 15, 2025 | 21:00 Hours

Location: TechNova Inc. HQ

Victim: The CEO

The CEO was found dead in their office. As the lead Data Analyst, you have been granted access to the company's secure SQL databases: Keycard Logs, Phone Records, and Alibi Statements.

Objective: Use SQL to reconstruct the crime and identify the killer.



| Step 1: The Crime Scene

📍 ESTABLISHING TIME & PLACE

We filter the evidence table to pinpoint the exact moment the crime was discovered in the CEO's office.

```
SELECT room, found_time
FROM evidence
WHERE room LIKE 'CEO Office'
ORDER BY found_time
LIMIT 1;
```

The 5 stages of a digital forensics investigation



2 - Preservation

Protect digital evidence from alteration or loss by creating a forensically sound image backup.

3 - Analysis

Meticulously examine collected data and artifacts to reconstruct the events surrounding the cyberattack.

4 - Documentation

Record all relevant evidence and findings in a clear, concise manner using professional formats.



| Step 2: Keycard Access Logs

SUSPECT LIST

We analyze the `keycard_logs` to identify who was physically present in the CEO's Office during the critical window (20:30 - 21:00).

```
SELECT e.name, e.role, kl.room,  
       kl.entry_time, kl.exit_time  
FROM keycard_logs kl  
INNER JOIN employees e  
  ON kl.employee_id = e.employee_id  
WHERE kl.room = 'CEO Office'  
      AND kl.entry_time BETWEEN  
        '2025-10-15 20:30:00'  
      AND '2025-10-15 21:00:00'  
ORDER BY kl.entry_time;
```

 Access log data stream

Step 3: Verifying Alibis

TRUTH VS. FICTION

We cross-reference the `alibis` table with the actual `keycard_logs`. A discrepancy between their claimed location and their badge scan reveals deception.

```
SELECT e.name, kl.employee_id,  
       kl.room AS actual_location,  
       a.claimed_location AS story_location,  
       a.claim_time  
FROM alibis a  
JOIN keycard_logs kl  
  ON a.employee_id = kl.employee_id  
INNER JOIN employees e  
  ON e.employee_id = a.employee_id  
WHERE a.claim_time BETWEEN  
       kl.entry_time AND kl.exit_time  
AND a.claimed_location ≠ kl.room;
```

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Step 4: Network Analysis



TRACING COMMUNICATIONS

We investigate the `calls` table to see who was communicating during the murder timeline (20:40 - 21:00).

```
SELECT e.name AS caller,  
       e2.name AS receiver,  
       c.caller_id, c.receiver_id,  
       c.call_time, c.duration_sec  
FROM calls c  
JOIN employees e  
  ON c.caller_id = e.employee_id  
JOIN employees e2  
  ON c.receiver_id = e2.employee_id  
WHERE c.call_time BETWEEN  
      '2025-10-15 20:40:00'  
      AND '2025-10-15 21:00:00';
```


Step 5: The Lie Detector

⚠️ AUTOMATED LOGIC

Using a CASE statement, we programmatically flag statements. If the claimed location matches the log, it's 'Truth'. Otherwise, 'LIE DETECTED'.

```
SELECT e.name, a.claim_time,  
       a.claimed_location AS what_they_said,  
       kl.room AS what_badge_logs_show,  
       CASE  
         WHEN a.claimed_location = kl.room THEN 'Truth'  
         ELSE 'LIE DETECTED'  
       END AS status  
FROM employees e  
JOIN alibis a ON e.employee_id = a.employee_id  
JOIN keycard_logs kl ON e.employee_id = kl.employee_id  
WHERE a.claim_time BETWEEN kl.entry_time AND kl.exit_time  
ORDER BY status ASC;
```

The 5 stages of a digital forensics investigation



| Step 6: Identifying the Killer

💀 THE FINAL INTERSECTION

We combine all clues: Presence in CEO Office + Suspicious Call Time + Falsified Alibi. The intersection of these datasets points to one person.

```
SELECT e.name AS killer
FROM employees e
JOIN keycard_logs kl ON e.employee_id = kl.employee_id
JOIN calls c ON e.employee_id = c.caller_id
JOIN alibis a ON e.employee_id = a.employee_id
WHERE kl.room = 'CEO Office'
      AND c.call_time BETWEEN
        '2025-10-15 20:40:00'
        AND '2025-10-15 21:00:00'
      AND a.claimed_location ≠ kl.room;
```



| Case Closed: Deliverables



SQL MASTERY

Demonstrated proficiency in JOINS, INTERSECTS, and Subqueries to solve complex logical problems.



THE VERDICT

The killer has been identified. Evidence has been secured and logged into the master database.



PROJECT STATUS

Challenge Completed.
21 Days SQL Challenge
Indian Data Club



| Image Sources



https://media.istockphoto.com/id/873894420/vector/detective-on-crime-scene.jpg?s=612x612&w=0&k=20&c=ZiC3OSTlwKn3kWgmOmZIGLZgw_jlIW3IAhLygYgwjRrY=

Source: www.istockphoto.com



<https://www.provenda.com/wp-content/uploads/2020/07/DIGITAL-FORENSICS-PROCESS-1024x647.png>

Source: www.provenda.com



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Source: www.aratek.co



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